

CSA5117 - Cryptography and Network Security

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Sub:- Assignment - 1

90
100
22/7/21

DEFEND THE WALL" - Encrypt key = 3

Cipher text :- GHIHQG WKH ZDOO

2. Decrypt "KHOOZ ZRUDG" key = 3

Plain text : HELLO WORLD

3. SECRET → VHFUHW key = 3

Decrypt WKLV LVDWHVW

THIS IS A TEST

Question - 2

1. Encrypt "ARE"

A → Q, B → W, C → E, D → R, E → T, F → Y

F → Y, I → O, R → K, E → T

Cipher text : YOKT

2. Frequency Analysis

Cipher text : UJHHD QEB NREFZH YOLTK

Best plain text : THE QUICK BROWN FOX

3. Question - 3

1. Encrypt

Plain text : ATTACKATPALE

Keyword : LEMON

Cipher text : LXFORVEFRNHR

2. Decrypt

Cipher text : LXFORVEFRNHR

Keyword : LEMON

Plain text : ATTACK AT DAWN

3.

Decrypt

Cipher text: QNXEPKMAEGITU

Keyword: KEY

Plain text: GIVEYOURBEST

Question - 4

Key matrix $\begin{bmatrix} 3 & 8 \\ 2 & 5 \end{bmatrix}$ encrypt "HI"

$$H = 7$$

$$I = 8$$

$$\begin{bmatrix} 3 & 8 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Modulo 26

$$45 \rightarrow 19 \text{ (T)}$$

$$54 \rightarrow 2 \text{ (C)}$$

Cipher text: TC

Verify :- PO $\rightarrow$ QC

verified using the matrix multiplication

Question - 5

Encrypt

Plain text: MEETMEATERLUNCH

Keyword: ZEBRAS

Cipher text: LIFKR8DJUIPCMOTI

Decrypt

Ciphertext : EAMTEFCUHTETREI

Keyword : BOARD

Plaintext : MEETMEATERNOON

Question - 6

Encrypt "HELLO" Using key "XMCKL"

Using $A=0, B=1, \dots, Z=25$.

Plaintext	H	E	L	L	O
Key	X	M	C	K	L
Cipher	E	Q	N	V	Z

Ciphertext : EQNVZ

2. Why is OTP Unbreakable.

⇒ Key is truly random

⇒ Key length equals message length

⇒ Key is used only once

⇒ The Key is kept secret

Question - 7

1. Encrypt "WE ARE DISCOVERED"

Keyword : ZEBRA

Remove Spaces : WEAREDISCOVERED.

Arrange :-

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Column Order (A, B, E, R, Z):

A → EDD
 B → ASR
 E → EIE
 R → RLE
 Z → WDV

Ciphertext: EDDASREIERCENDV

2. Decrypt

Ciphertext: EVLNEALPTKESAEAROFDSDEECUXWN

Keyword: TRIAL

Plaintext: WEAREDISCOVEREDFLEET ONCE

3. Information Security Using Keyword "HACK"

Column - 4

Rows needed - 5

Padding required - x

Encrypted text:

ITYNEUDFSARITICORMNEXX

Text :-

ZQSDVUDHXMDPVGIPOZPEVSGZW SZDPFPESXUZDETUCG

Letter Frequency :-

Z - Most Frequent

U - High

D - High

P - High

Problem Mapping

Z → E

U → T

D → A

P → O

Partial Decryption :-

Some words become recognisable after substitutions

Best Guess :-

"THIS IS A SECRET MESSAGE"

Question-9

Ciphertext :

B JTWD JWYN JWNXFXNRUDJWX

Known plaintext :

THIS IS THE

Mapping :-

B → T

J → H

W → I

Q → S

Y → E

N → T

Decrypted Message.

THIS IS THE SIMPLEST

Question - 10

Ciphertext :

ZICVTWQNGIRZGVTWAVZHCOYGLMGJ

Recovered key

Key plain text :-

CRYPTOGRAPHY IS FUN AND USEFUL